

USER INTERFACE DESIGN

EXPERIMENT 9: Design Input Forms with Validation using HTML, CSS and JavaScript

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AIM:

To design and develop a **user registration form** that validates multiple user inputs such as name, email, phone number, and password using HTML, CSS, and JavaScript.

PROCEDURE:

Step 1: Creating the HTML Structure

A registration form was created with the following input fields:

- Name
- Email Address
- Phone Number
- Password
- Submit Button

Each input field is associated with a `<span>` element to display validation error messages.

Step 2: Styling using CSS

The form was styled to enhance user experience:

- Centered layout using flexbox
- Gradient background for visual appeal
- Rounded input fields and container
- Styled submit button with hover effect
- Error messages displayed in red
- Success message displayed in green

Step 3: Implementing JavaScript Validation

JavaScript was used to validate user inputs when the form is submitted.

Name Validation:

- Ensures the field is not empty

Email Validation:

- Checks for proper email format using pattern matching

Phone Number Validation:

- Ensures input contains exactly 10 digits
- Allows only numeric values

Password Validation:

- Must contain at least 6 characters

Step 4: Displaying Validation Messages

- If any field is invalid, corresponding error messages are displayed below the input field
- If all inputs are valid, a success message is displayed

CODE:

UID.HTMl:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Registration Form</title>
<link rel="stylesheet" href="style.css">
</head>

<body>
```

```

<div class="container">
<form id="myForm">

<h2>User Registration</h2>

<label>Name:</label>
<input type="text" id="name" placeholder="Enter your name">
<span id="nameError" class="error"></span>

<label>Email:</label>
<input type="text" id="email" placeholder="Enter your email">
<span id="emailError" class="error"></span>

<label>Phone Number:</label>
<input type="text" id="phone" placeholder="Enter phone number">
<span id="phoneError" class="error"></span>

<label>Password:</label>
<input type="password" id="password" placeholder="Enter password">
<span id="passwordError" class="error"></span>

<button type="submit">Register</button>

<p id="success"></p>

</form>
</div>

<script src="script.js"></script>

</body>
</html>

```

STYLE.CSS:

```

body{
  margin:0;
  padding:0;
  font-family:Arial, sans-serif;
  background: linear-gradient(to right, #141e30, #243b55);

```

```
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
}

.container{
    background:white;
    padding:30px;
    width:350px;
    border-radius:15px;
    box-shadow:0 8px 20px rgba(0,0,0,0.3);
}

h2{
    text-align:center;
    margin-bottom:20px;
    color:#333;
}

form{
    display:flex;
    flex-direction:column;
}

label{
    margin-top:10px;
    margin-bottom:5px;
    font-weight:bold;
    color:#444;
}

input{
    padding:12px;
    border:1px solid #ccc;
    border-radius:8px;
    font-size:15px;
}

input:focus{
```

```
    outline:none;
    border-color:#6c63ff;
}

button{
    margin-top:20px;
    padding:12px;
    background:#6c63ff;
    color:white;
    border:none;
    border-radius:8px;
    font-size:16px;
    cursor:pointer;
}

button:hover{
    background:#5848e5;
}

.error{
    color:red;
    font-size:13px;
    margin-top:3px;
}

#success{
    color:green;
    text-align:center;
    margin-top:15px;
    font-weight:bold;
}
```

SCRIPT.JS:

```
document.getElementById("myForm").addEventListener("submit", function(e){

    e.preventDefault();

    // Get values
    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
```

```
let phone = document.getElementById("phone").value.trim();
let password = document.getElementById("password").value.trim();

// Patterns
let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
let phonePattern = /^[0-9]{10}$/;

// Clear previous errors
document.getElementById("nameError").textContent="";
document.getElementById("emailError").textContent="";
document.getElementById("phoneError").textContent="";
document.getElementById("passwordError").textContent="";
document.getElementById("success").textContent="";

// Validation flags
let isValid = true;

// Name validation
if(name === ""){
    document.getElementById("nameError").textContent="Name is required";
    isValid = false;
}

// Email validation
if(!emailPattern.test(email)){
    document.getElementById("emailError").textContent="Enter a valid email";
    isValid = false;
}

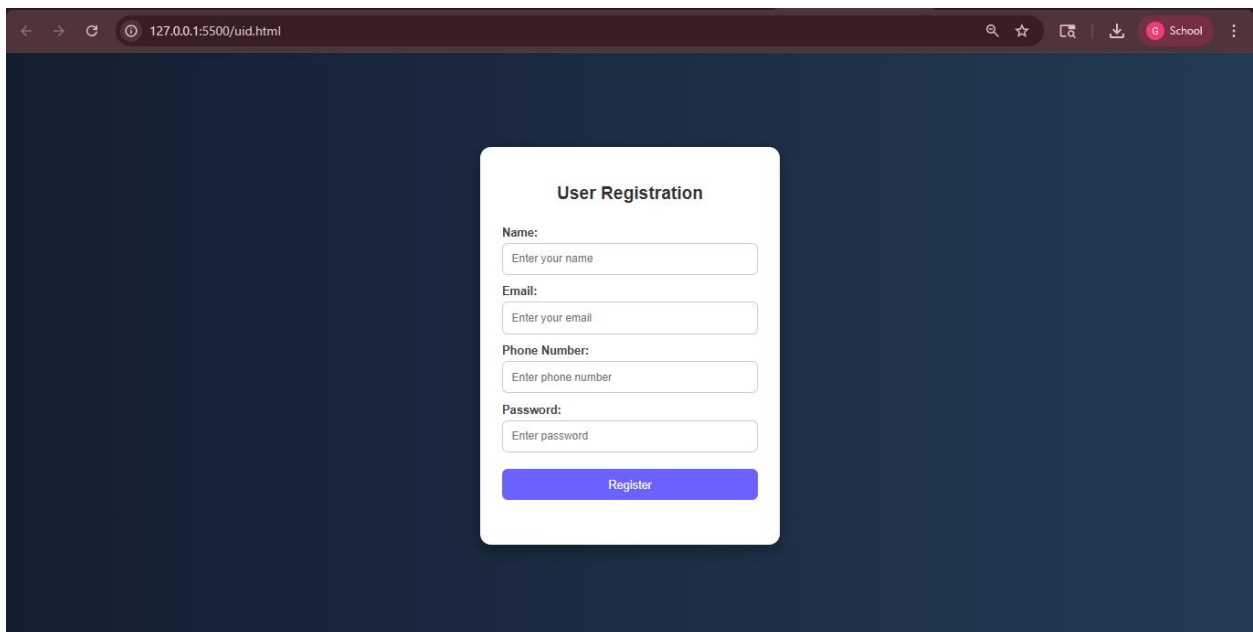
// Phone validation
if(!phonePattern.test(phone)){
    document.getElementById("phoneError").textContent="Enter 10 digit phone number";
    isValid = false;
}

// Password validation
if(password.length < 6){
```

```
    document.getElementById("passwordError").textContent="Minimum 6
characters required";
    isValid = false;
}

// Success
if(isValid){
    document.getElementById("success").textContent="Registration
Successful!";
}
});
```

OUTPUT:



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/uid.html". The browser's toolbar includes navigation buttons (back, forward, refresh), a search icon, a star icon, a print icon, a download icon, and a tab labeled "School". The main content area features a dark blue background with a white registration form centered. The form is titled "User Registration" and contains four input fields with placeholder text: "Enter your name", "Enter your email", "Enter phone number", and "Enter password". Below these fields is a blue "Register" button.

The image displays two sequential screenshots of a web browser window showing a "User Registration" form. The browser's address bar indicates the URL "127.0.0.1:5500/uid.html".

Top Screenshot (Initial State):

- Title:** User Registration
- Name:** Input field containing "Gayu".
- Email:** Input field containing "gayathri32110gmail.com". A red error message "Enter a valid email" is displayed below the field.
- Phone Number:** Input field containing "978609731". A red error message "Enter 10 digit phone number" is displayed below the field.
- Password:** Input field containing "****". A red error message "Minimum 6 characters required" is displayed below the field.
- Action:** A blue "Register" button is located at the bottom of the form.

Bottom Screenshot (Successful State):

- Title:** User Registration
- Name:** Input field containing "Gayu".
- Email:** Input field containing "gayathri32110@gmail.com".
- Phone Number:** Input field containing "9786097310".
- Password:** Input field containing "*****".
- Action:** A blue "Register" button is located at the bottom of the form.
- Feedback:** A green message "Registration Successful!" is displayed below the "Register" button.

PROTOTYPE LINK:

<http://127.0.0.1:5500/uid.html>

RESULT:

A responsive and user-friendly registration form was successfully designed. The form validates user inputs effectively and provides appropriate feedback messages, ensuring correct data entry.

